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Oefeningenreeks 6E nr. 8.2

Bereken $\cos\left(\frac{1}{3}\right)$ tot op 12 decimalen door middel van een Taylorreeks. Geef ook de restterm.

1. Taylorontwikkeling van $\cos x$ rond 0

   $f(x) = \cos x$

   $f'(x) = -\sin x$

   $f''(x) = -\cos x$

   $f'''(x) = \sin x$

   $f^{iv}(x) = \cos x$

   $f(0) = 1$

   $f'(0) = 0$

   $f''(0) = -1$

   $f'''(0) = 0$

   $f^{iv}(0) = 1$

   $\Rightarrow f^{(2k+1)}(0) = 0$ en $f^{(2k)}(0) = (-1)^k$

   $$\Rightarrow T_n\left(\cos x, 0\right)(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

   Alleen even termen blijven over:

   $$= \sum_{k=0}^{\infty} \frac{f^{(2k)}(0)}{(2k)!} x^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$$

   Dus

   $$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + R_{10}(x)$$

Stel nu $x = \dfrac{1}{3}$, dan hebben we dus

$$\begin{aligned}
\cos\left(\frac{1}{3}\right) &= 1 - \frac{\left(\frac{1}{3}\right)^2}{2!} + \frac{\left(\frac{1}{3}\right)^4}{4!} - \frac{\left(\frac{1}{3}\right)^6}{6!} + \frac{\left(\frac{1}{3}\right)^8}{8!} - \frac{\left(\frac{1}{3}\right)^{10}}{10!} + R_{10}\left(\frac{1}{3}\right) \\
&\approx 0.944956946314085 + R_{10}\left(\frac{1}{3}\right)
\end{aligned}$$

2. Restterm

$$R_{10}\left(\frac{1}{3}\right) = \frac{f^{(12)}(\xi)}{12!}\left(\frac{1}{3}\right)^{12} \qquad \xi \in \left[0, \frac{1}{3}\right]$$

$$\left|f^{(12)}(\xi)\right| \leq 1$$

$$\Rightarrow \left|R_{10}\left(\frac{1}{3}\right)\right| \leq \frac{\left(\frac{1}{3}\right)^{12}}{12!}$$

$$\Rightarrow \left|R_{10}\left(\frac{1}{3}\right)\right| = \frac{1}{3^{12} \cdot 12!} \approx 3.928206751 \cdot 10^{-15}$$

Aangezien

$$3.928206751 \cdot 10^{-15} < 10^{-12}$$

mogen we afronden tot op 12 decimalen. Bijgevolg is

$$\cos\left(\frac{1}{3}\right) \approx 0.944956946315$$

References